

STUDENT NO : _____

Class: _____



KILLARA HIGH SCHOOL

MATHEMATICS

HSC ASSESSMENT - TASK 1

Year 11 Term 4, 30-11-2011

Time allowed: 1 hour

Maximum marks: 54 marks

INSTRUCTIONS:

- Approved calculators may be used.
- All questions are compulsory and should be attempted.
- Use blue or black biro, pencil for graphs and diagrams.
- Where appropriate, FULL WORKING must be shown.
- Diagrams are NOT TO SCALE.

Format:

Part A: Multiple choice questions (6 marks)

Answer on multiple choice answer sheet

Part B: Four questions (48 marks)

Answer with full working on paper provided.

Start a new sheet for each question.

PART A – Multiple Choice questions (1 mark each)

1. Find the equation of the parabola with coordinates of the vertex $(0, 0)$, equation of the axis $y = 0$ and focal length 9.

(A) $x^2 = \pm 36y$ (B) $x^2 = -12y$ (C) $y^2 = \pm 36x$ (D) $y^2 = -12x$

2. The focus of the parabola $(x-1)^2 = 8(y-2)$ is at

(A) $(0, 2)$ (B) $(1, 2)$ (C) $(1, 0)$ (D) $(1, 4)$

3. If α and β are the roots of $x^2 - 5x - 3 = 0$ then

(A) $\alpha + \beta = 5$ and $\alpha\beta = -3$ (B) $\alpha + \beta = -5$ and $\alpha\beta = 3$

(C) $\alpha + \beta = -5$ and $\alpha\beta = -3$ (D) $\alpha + \beta = 5$ and $\alpha\beta = 3$

4. If $Px^2 + (P+Q)x + (P+R) \equiv 9x^2 + 7x + 2$ then

(A) $P = 9, Q = 7$ and $R = 2$ (B) $P = 9, Q = 2$ and $R = 7$

(C) $P = 9, Q = -7$ and $R = -2$ (D) $P = 9, Q = -2$ and $R = -7$

5. The fortieth term of the series for which $T_n = 5n - 9$ is?

(A) 25 (B) 191 (C) 155 (D) 61

6. The number of terms in the series $1 + 1.0825 + 1.0825^2 + \dots + 1.0825^{27} + 1.0825^{28}$ is?

(A) 5 (B) 27 (C) 28 (D) 29

PART B

Question 1 (12 marks)

a. Consider the parabola with equation $x^2 = 4(y-1)$.

- (i) Find the coordinate of the vertex of the parabola. 1
- (ii) Find the coordinate of the focus of the parabola. 1

b. Show that

$x^2 - 2x + y^2 + 4y + 1 = 0$ and $x^2 - 2x + y^2 + 4y - 4 = 0$ are concentric
(have the same centre). 2

c. Let $R\left(\frac{1}{5}, -2\right)$ be a point on the parabola $y^2 = 20x$.

- (i) Show the equation of the focal chord passing through R
is $5x - 12y - 25 = 0$ 2
- (ii) Find the coordinates of the point Q where this chord
cuts the directrix. 2
- (iii) Find the perpendicular distance from the focal chord
to the point $P(-1, -7)$. 2
- (iii) Hence find the area of ΔPQR . Or use alternate methods. 2

Question 2 (12 marks)

- a. Find the values of k for which the quadratic equation $3x^2 + 2x + k = 0$ has no real roots. 2
- b. Let α and β be the solutions of $x^2 - 3x + 1 = 0$. Find
- (i) $\frac{1}{\alpha} + \frac{1}{\beta}$ 1
- (ii) $\alpha^2 + \beta^2$. 1
- c. Prove that $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$ 2
- d. Find values of k in the equation $x^2 + (k+1)x + (\frac{k+1}{4}) = 0$ if:
- (i) roots are equal in magnitude but opposite in sign 1
- (ii) one root is 1 1
- e. (i) Show that the x - coordinates of the points of intersection of a circle $x^2 + y^2 = 4$ and the line $y = x + 1$ satisfy the equation $2x^2 + 2x - 3 = 0$. 2
- (ii) Evaluate the discriminant Δ and explain why this shows that there are two points of intersection. 2

Question 3 (12 marks)

- a. The first term of an arithmetic series is 15 and the third term is 31. Find the

- (i) Common difference 1
- (ii) 10^{th} term of the series. 1

- b. Evaluate $\sum_3^{17} 5n - 6$. 2

- c. (i) Show that $(x+1) + (2x+4) + (3x+7) + \dots$ is an arithmetic series. 1

- (ii) Find the sum of the first 50 terms of the series, in terms of x . 1

- d. The 20^{th} term of an arithmetic series is 131 and the sum of the 6^{th} to 10^{th} terms inclusive is 235. Find first 3 terms. 2

- e. For what value of n is the sum of the arithmetic series $5 + 9 + 13 + \dots$ equal to 152? 2

- f. The n^{th} term of an arithmetic sequence is $T_n = 7 + 4n$.
Prove that $5T_1 + 4T_2 = T_{27}$. 2

Question 4 (12 marks)

- a. Which term of the series $8 - 4 + 2 - \dots$ is equal to $\frac{1}{128}$? 2
- b. The common ratio of a geometric series is 4 and the sum of the first 5 terms is 3069. Find the first term. 2
- c. A bouncy ball drops from a height of 9 metres and bounces continually, each successive height being $\frac{2}{3}$ of the previous height.
- (i) Show that the first distance travelled down-and-up is 15 metres, and show that the successive down-and-up distances form a geometric series. 2
- (ii) Through what distance does the ball eventually travel? 1
- d. Consider the geometric series $1 + (\sqrt{5} - 2) + (\sqrt{5} - 2)^2 + \dots$
- (i) Explain why the geometric series has a limiting sum. 1
- (ii) Find the exact value of the limiting sum. Write your answer with a rational denominator. 2
- e. Pam deposited \$2000 into an account that earned interest monthly. For 2 years the interest rate was 12% p.a., then it fell to 7% p.a. How much money will there be in Pam's account after 5 years? 2

Part A Answer Sheet

STUDENT NO : _____

Total marks: 6

Attempt question 1 – 6

Select the alternative A, B, C and D that best answers the question and indicate your choice by shading the ellipse.

1 A ☐ B ☐ C ☐ D ☐

2 A ☐ B ☐ C ☐ D ☐

3 A ☐ B ☐ C ☐ D ☐

4 A ☐ B ☐ C ☐ D ☐

5 A ☐ B ☐ C ☐ D ☐

6 A ☐ B ☐ C ☐ D ☐

Part A. Multiple choice: 1.(C) 2.(D) 3.(A) 4.(D) 5.(B) 6.(D)

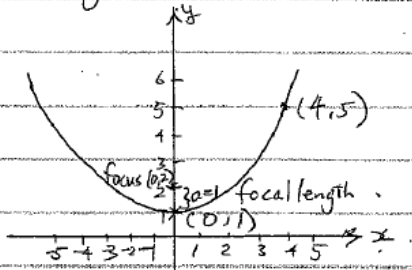
Part B (1)

(i) $x^2 = 4(y-1)$, $(x-h)^2 = 4a(y-k)$

giving $(x-0)^2 = 4 \times 1 \times (y-1)$

(i) The vertex is at $(h, k) = (0, 1)$ ✓

(ii) Focal length = 1 unit ($\because a = 1$)



From the diagram the focus is at $(0, 2)$

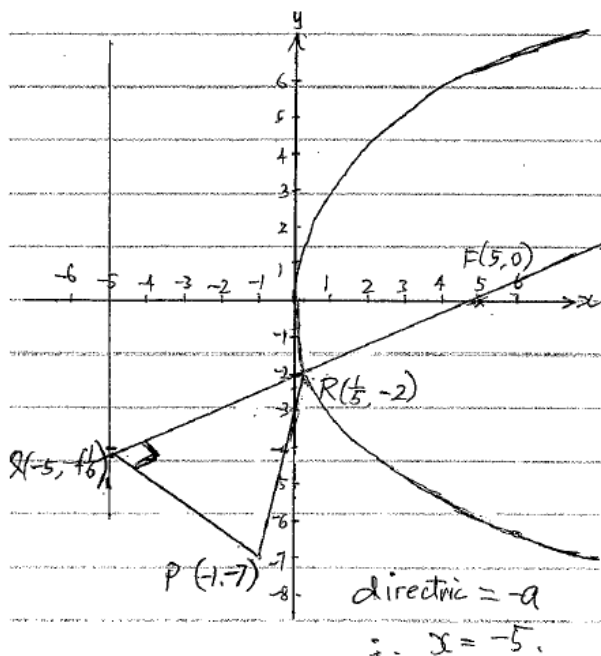
(b) $x^2 - 2x + 1 + y^2 + 4y + 4 = 1 + 1 + 4$ & $x^2 - 2x + 1 + y^2 + 4y + 4 = 4 + 4 + 1$

$(x-1)^2 + (y+2)^2 = 2^2$ & $(x-1)^2 + (y+2)^2 = 3^2$

$x=1$ & $y=-2$ radius=2 & $x=1$, $y=-2$ radius=3

$\therefore$ Centre $(1, -2)$ radius=2 & centre $(1, -2)$ radius=3

(a) $R(\frac{1}{5}, -2)$ $y^2 = 20x$



(i) $4a = 20$ $a = \frac{20}{4} = 5$

$\therefore$ focus $(5, 0)$

Gradient = $\frac{-2-0}{\frac{1}{5}-5} = \frac{-2}{-\frac{24}{5}} = \frac{5}{12}$

$\therefore$ equation of line

$y-0 = \frac{5}{12}(x-5)$

$12y = 5x - 25$

$5x - 12y - 25 = 0$ ✓

(ii) directrix = $-a$ $\therefore x = -5$

$12y - 5(-5) + 25 = 0$ ✓

$12y = -50$

$y = -4\frac{1}{6}$ $\therefore (-5, -4\frac{1}{6})$ ✓

(iii) $P(-1, -7)$ distance of PQ

$\therefore$ dist of PQ = $\frac{1-5(-1)+12(-7)+25}{\sqrt{5^2+12^2}}$ ✓

$= \frac{54}{13}$ units ✓

(iv) Find the area of $\triangle PQR$.

$R(\frac{1}{5}, -2)$ $Q(-5, -4\frac{1}{6})$

PQ = $\frac{54}{13}$ units from Q to P.

$QR = \sqrt{(-5-\frac{1}{5})^2 + (-4\frac{1}{6}-2)^2}$
 $= \frac{169}{30}$ ✓

$\therefore$ Area of $\triangle PQR = \frac{1}{2} \times (PQ \times QR)$

$= \frac{1}{2} \times \frac{54}{13} \times \frac{169}{30}$

$= 11.7$ units² ✓

$$2(a) \quad 3x^2 + 2x + k = 0, \quad \Delta = b^2 - 4ac$$

$$\Delta < 0 \quad \text{So } \Delta = 4 - 4 \times 3 \times k \\ = 4 - 12k \quad \checkmark$$

For no real roots $\Delta < 0$

$$\therefore 4 - 12k < 0$$

$$12k > 4$$

$$k > \frac{1}{3} \quad \checkmark$$

$$(b) \quad x^2 - 3x + 1 = 0$$

$$\alpha + \beta = -\frac{b}{a} = -\frac{-3}{1} = 3$$

$$\alpha\beta = \frac{c}{a} = \frac{1}{1} = 1$$

$$(i) \quad \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$$

$$= \frac{3}{1}$$

$$= 3 \quad \checkmark$$

$$(ii) \quad \alpha^3 + \beta^3$$

$$(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$

$$= (-3)^3 - 3(1)$$

$$= 9 - 2$$

$$= 7 \quad \checkmark$$

$$(c) \quad \text{Prove that } (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

Proof:

$$\text{LHS } (\alpha - \beta)^2 = (\alpha - \beta)(\alpha - \beta)$$

$$= \alpha^2 - 2\alpha\beta + \beta^2 \quad \checkmark$$

$$= \alpha^2 + 2\alpha\beta + \beta^2 - 4\alpha\beta$$

$$= (\alpha + \beta)^2 - 4\alpha\beta$$

$$= \text{RHS}$$

$$(d) \quad x^2 + (k+1)x + \left(\frac{k+1}{4}\right) = 0$$

$$(i) \quad \alpha + \beta = 0 \quad \alpha - \alpha = -\frac{(k+1)}{4}$$

$$\text{So } -(k+1) = 0$$

$$-k - 1 = 0$$

$$-k = 1$$

$$k = -1 \quad \checkmark$$

$$(ii) \quad x = 1$$

$$\text{So } 1 + k + 1 + \frac{k+1}{4} = 0$$

$$4 + 4k + 4 + k + 1 = 0$$

$$5k = -9$$

$$k = -\frac{9}{5}$$

$$k = -1.8 \quad \checkmark$$

$$(e) (i) \quad x^2 + y^2 = 4 \quad \& \quad y = x + 1$$

$$y^2 = 4 - x^2$$

$$y^2 = (x+1)^2$$

$$4 - x^2 = (x+1)^2 \quad \checkmark$$

$$4 - x^2 = x^2 + 2x + 1$$

$$2x^2 + 2x - 3 = 0 \quad \checkmark \text{ is}$$

satisfy the equation.

$$\Delta = b^2 - 4ac$$

$$(ii) \quad = 2^2 - 4 \times 2 \times (-3)$$

$$= 4 + 24$$

$$= 28 \quad \checkmark$$

Since $\Delta > 0$, the quadratic equation has two roots means there are two points of intersection.

$$(3)(a) T_n = a + (n-1)d \quad a = 15$$

$$(i) T_1 = a + 0d = 15$$

$$T_3 = a + 2d = 31$$

$$2d = 16$$

$$d = 8 \quad \checkmark$$

(ii) 10th term of the series

$$T_{10} = 15 + (10-1)8$$

$$= 15 + 9 \times 8$$

$$= 87 \quad \checkmark$$

$$(b) \sum_{n=3}^{17} 5n - 6$$

$$n=3, \therefore 5(3) - 6 = 9$$

$$n=4, \quad 5(4) - 6 = 14$$

$$n=5, \quad 5(5) - 6 = 19$$

$$\therefore a = 9 \quad d = 5 \quad \checkmark$$

$$n=17, \quad 5(17) - 6 = 79$$

$$T_{15} = \frac{15}{2} (9 + 79)$$

$$= 660 \quad \checkmark$$

$$(c)(i) (x+1) + (2x+4) + (3x+7) + \dots$$

$$d = (2x+4) - (x+1) = x+3$$

$$(3x+7) - (2x+4) = x+3$$

common difference equalled each other

$$\therefore (x+1) + (2x+4) + (3x+7) + \dots$$

is an arithmetic Series. $\checkmark$

$$(ii) S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{50} = \frac{50}{2} [2(x+1) + 49(x+3)]$$

$$= 25 [2x+2 + 49x+147]$$

$$= 25 [51x + 149]$$

$$= 1275x + 3725 \quad \checkmark$$

(d)

$$T_{20} = a + 19d = 131$$

$$S_{10} = \frac{10}{2} [2a + (9)d]$$

$$= 10a + 45d$$

$$S_5 = \frac{5}{2} [2a + 4d]$$

$$= 5a + 10d$$

$$\therefore (10a + 45d) - (5a + 10d) = 235$$

$$5a + 35d = 235 \quad \checkmark$$

$$a + 7d = 47 \quad \text{--- (1)}$$

$$a + 19d = 131 \quad \text{--- (2)}$$

$$(2) - (1) \quad 12d = 84$$

$$d = 7$$

$$\therefore a + 49 = 47$$

$$a = -2 \quad \checkmark$$

First 3 terms $-2, 5, 12, \dots$

$$5 + 9 + 13 + \dots + = 152$$

$$(c) S_n = \frac{n}{2} [2a + (n-1)d]$$

$$152 = \frac{n}{2} [10 + (n-1)4]$$

$$152 = \frac{n}{2} [4n + 6] \quad \checkmark$$

$$152 = n(2n + 3)$$

$$2n^2 + 3n - 152 = 0$$

$$(2n - 19)(n - 8) = 0$$

$$\therefore n = 8$$

$$\therefore T_n = 7 + 4n$$

Prove that $5T_1 + 4T_2 = T_{27}$.

$$\text{LHS } 5(7+4) + 4(7+8) = 7 + 4(27)$$

$$35 + 20 + 28 + 32 = 7 + 108$$

$$115 = 115 \quad \checkmark$$

$$= \text{RHS}$$

$$\therefore 5T_1 + 4T_2 = T_{27}$$

$$\text{LHS} = \text{RHS}$$

(4) (a)

$$8 - 4 + 2 - \dots = \frac{1}{128}$$

$$a = 8, \quad r = \frac{-4}{8} = \frac{2}{-4} = -\frac{1}{2}$$

$$T_n = ar^{n-1}$$

$$8\left(-\frac{1}{2}\right)^{n-1} = \frac{1}{128}$$

$$\left(-\frac{1}{2}\right)^{n-1} = \frac{1}{1024} \quad \checkmark$$

$$(-1)^{n-1} \left(\frac{1}{2}\right)^{n-1} = \left(\frac{1}{2}\right)^{10}$$

$$n-1 = 10$$

$$n = \underline{\underline{11}} \quad \checkmark$$

(b) $r = 4$ $a = ?$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_5 = \frac{a(4^5 - 1)}{4 - 1} = 3069 \quad \checkmark$$

$$a(1023) = 3(3069)$$

$$a = \frac{9207}{1023}$$

$$\text{show } a = \underline{\underline{9}} \quad \checkmark$$

(c) (i) 1st distance = $9 + 9\frac{2}{3} = 9 + 6 = 15\text{m}$

The successive down-and-up distances

form a GP with $a = 15$ and $r = \frac{2}{3}$ $\checkmark$

$$(ii) S_{\infty} = \frac{a}{1-r}$$

$$= \frac{15}{1 - \frac{2}{3}}$$

$$= \frac{15}{\frac{1}{3}}$$

$$= \underline{\underline{45}} \text{ metres } \checkmark$$

4(b)

$$1 + (\sqrt{5}-2) + (\sqrt{5}-2)^2 + \dots$$

(i) Limiting sum exists when $|r| < 1$

$$r = \sqrt{5}-2, \quad a = 1$$

$$\therefore |r| = |\sqrt{5}-2| = 0.236\dots < 1 \quad \checkmark$$

$\therefore$ Limiting sum exists.

$$(ii) S_{\infty} = \frac{a}{1-r}$$

$$= \frac{1}{1 - (\sqrt{5}-2)} \quad \checkmark$$

$$= \frac{1}{3-\sqrt{5}} \quad \checkmark$$

$$n = 2 \times 12 = 24 \text{ months}$$

$$n = 3 \times 12 = 36 \text{ months}$$

$$(e) A = 2000 \left(1 + \frac{1}{100}\right)^{24} + (2539.4692) \left(1 + \frac{7}{100}\right)^{12}$$

1st 2 yrs was 12%

After 2nd yr interest fell to 7% p.a.

$$= \underline{\underline{\$3130.98}} \quad \checkmark$$

$$A = P \left(1 + \frac{r}{100}\right)^n$$

